

Tier-Prime-Lifetime: Density-Band Lifetime Formulas

A Full Formulation of Structure Survival Time Across the UST Scale Hierarchy

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Abstract

Unified Substrate Theory (UST) is a substrate-geometry framework in which every physical entity — particles, atoms, organisms, stars, galaxies, and the universe itself — is a structured frequency pattern riding on a universal substrate field. The substrate is the medium that underlies all matter and energy. It is not either of those things — it carries both. Within UST, the prime-tier hierarchy defines the natural scaffold of physical scale: thirteen prime numbers (2 through 41) label discrete resonance levels of the substrate, from the Planck floor at Tier 2 to the observable universe at Tier 41. Each tier corresponds to a characteristic physical length scale and the class of structures that form there.

The central measure of structural stress in UST is the density ratio D_r — the ratio of a structure's local mass-energy density to the critical density at its prime tier. When $D_r = 1.0$, the structure sits at perfect equilibrium with the substrate and is, in principle, indefinitely stable. Deviations in either direction — underdense (D_r less than 1) or overdense (D_r greater than 1) — stress the frequency lock and drive decay or collapse. This paper derives and applies the core lifetime formula $L(p, D_r) = \tau_{\text{base}}(p) \times p$ to the power α divided by the absolute deviation of D_r from 1.0 raised to $\beta(\text{band})$, computing predicted structure survival times for all 13 prime tiers across five density bands: Ghost (D_r less than 0.01), Quiet ($D_r = 0.01$ to 0.10), Active ($D_r = 0.10$ to 10.0), Hot ($D_r = 10.0$ to 1,000.0), and Critical (D_r greater than 1,000.0). Each band represents a physically distinct substrate stress regime, from near-void dispersion in the Ghost band to pre-collapse compression in the Critical band.

The key result is unambiguous: lifetime drops sharply as D_r moves far from 1.0 in either direction, and lifetime rises with tier prime number via the golden-ratio exponent $\alpha = 1.618$. The computed lifetimes span 62 orders of magnitude — from approximately 10^{-51} seconds at the Planck-floor Critical band to 2.2 trillion years at the universe-scale Active band. The state $UBL = 1$ — where the Universal Baseline Level ratio reaches unity — is simultaneously the fully-quiet substrate state, the black hole interior, and the pre-bounce threshold. It is the floor below all structure and the ceiling toward which Critical-band structures inevitably fall.

Section 1: What This Paper Is About

This paper does one thing. It calculates how long structures at each prime tier survive given their local density band. It is not speculative. It is a direct application of UST substrate mechanics using the density ratio, the tier prime number, and a five-band classification of local density relative to the critical density at that tier.

1.1 What a Tier Is

In UST, a "tier" is a resonance level of the substrate, labeled by a prime number. The substrate vibrates at fundamental prime frequencies. Structures form by locking into one of these resonance levels — a process called frequency lock. Primes are used because they are the irreducible resonance frequencies of the substrate. Composite numbers fold back to prime fundamentals. You cannot build a stable standing wave at a composite frequency without it decomposing into its prime factors. So the scaffold of scale is prime-numbered by necessity, not convention.

1.2 What Lifetime Means Here

Lifetime in this paper means the duration a coherent structure holds its frequency lock before density imbalance breaks it apart or collapses it. It is not a biological lifespan. It is not a geological age. It is the decay time constant of the resonance pattern at the given density ratio. Structures near $D_r = 1.0$ maintain their lock almost indefinitely by continuously feeding energy into the resonance — they stay alive by staying balanced. Structures pushed far from $D_r = 1.0$ bleed coherence and eventually dissolve or collapse.

1.3 The UBL = 1 State

UBL is the Universal Baseline Level ratio. At $UBL = 1$, the substrate has organized all of its available energy into coherent structures — nothing is left free. This triple-identity state is simultaneously: the black hole interior (maximum gravitational compression, zero free fluctuation), the fully-quiet substrate state (no substrate noise, no free wave modes), and the pre-bounce condition (the point at which the system must release energy outward). Structures cannot exist at $UBL = 1$ because they would be indistinguishable from the substrate itself — there would be no contrast, no frequency differential, no pattern to identify. $UBL = 1$ is the absolute floor of structure and the gateway to the next expansion cycle.

1.4 Why Density Matters

The density ratio D_r = local density divided by critical density at the tier is the single most important number for determining a structure's fate. When $D_r = 1.0$, the structure is at equilibrium — its inward self-gravity and outward substrate pressure are balanced, and the frequency lock is stable. When D_r falls below 1.0, substrate pressure wins and the structure disperses slowly into the surrounding field. When D_r rises above 1.0, internal density wins and the structure is being actively compressed inward. The further D_r sits from 1.0 in either direction, the faster the structure loses its coherence.

Section 2: Core UST Framework

2.1 The Substrate

The substrate is the universal field that underlies all structure. It is not matter. It is not energy. It is the medium that carries both. All particles, fields, atoms, stars, and galaxies are frequency patterns — standing wave structures — that form when the substrate organizes into stable resonance loops. The substrate itself has no preferred scale. It supports structure at every tier simultaneously. What differs between tiers is the characteristic wavelength of the resonance and the timescale over which imbalances propagate.

2.2 The Universal Balance Law (UBL)

UBL is the ratio of the substrate's organized energy to its total available capacity. It measures how much of the substrate's potential has already been locked into structure.

$$\text{UBL} = E_{\text{organized}} / E_{\text{total}}$$

UBL equals the energy currently locked into coherent structures divided by the maximum energy capacity of the substrate.

Symbol Key:

- **UBL** — Universal Balance Law ratio. Dimensionless. Range: 0 to 1.
- **E_organized** — Energy currently locked into coherent, structured frequency patterns (matter, fields, bound states).
- **E_total** — Maximum substrate energy capacity — the total available energy if every substrate mode were excited to its resonance peak.

When $UBL = 1$: all energy is organized. This is the black hole state, the fully-quiet substrate, and the bounce condition simultaneously. No free fluctuations exist. No structure can be distinguished from the medium itself. The system must release.

When UBL approaches 0: the substrate is at maximum disorder — maximum possible fluctuation. This is the Big Bang or inflationary expansion phase, where essentially no energy is yet locked into stable structure and the substrate is churning with free wave modes.

2.3 The Density Ratio D_r

At any given prime tier p , the local density compared to the critical density at that tier defines D_r .

$$D_r = \rho_{\text{local}} / \rho_{\text{crit}}(p)$$

Plain English: D_r equals the measured local mass-energy density at the structure's location divided by the critical density threshold specific to prime tier p .

Symbol Key:

- **D_r** — Density ratio. Dimensionless. The primary measure of substrate stress at a structure's location.
- **rho_{local}** — Measured local mass-energy density at the structure's location, in units appropriate to the tier (kg/m³ or equivalent).
- **rho_{crit(p)}** — Critical density threshold specific to prime tier p. This is the equilibrium density for structures resonating at tier p — the density at which the structure's self-gravity exactly balances the outward substrate pressure at that scale.

D_r = 1.0: Perfect equilibrium. The structure's local density matches its tier's critical density. Maximum structural stability — the structure is balanced against substrate pressure and can theoretically persist without degradation.

D_r less than 1.0: The structure is underdense. Substrate pressure exceeds local density. The structure is being stretched and will eventually disperse into the surrounding field.

D_r greater than 1.0: The structure is overdense. Local density exceeds equilibrium. The substrate is being compressed inward. Structures in this band collapse faster as D_r increases.

2.4 The Prime Tier Hierarchy

UST identifies thirteen prime tiers spanning from the Planck floor to the observable universe. Each tier is labeled by its prime number, has a characteristic physical length scale, and governs a specific class of physical structures.

Tier	Prime (p)	Characteristic Scale (m)	Physical Domain
T2	2	~10 ⁻³⁵	Planck floor, raw substrate lattice, pre-particle vacuum structure
T3	3	~10 ⁻³⁰	Quantum foam, sub-Planck fluctuation structures, pre-field condensates
T5	5	~10 ⁻¹⁸	Fundamental particle formation, quark-level resonance loops

Tier	Prime (p)	Characteristic Scale (m)	Physical Domain
T7	7	$\sim 10^{-15}$	Nuclear scale, hadron binding, proton/neutron structure
T11	11	$\sim 10^{-10}$	Atomic scale, electron shell resonance, chemical identity
T13	13	$\sim 10^{-9}$	Molecular scale, chemical bonds, nanoscale structure
T17	17	$\sim 10^{-6}$	Cellular/biological scale, organelle geometry, microbial structure
T19	19	$\sim 10^0$	Organism scale, macro-biological coherence, integrated living systems
T23	23	$\sim 10^6$	Planetary scale, crustal dynamics, atmospheric coherence
T29	29	$\sim 10^9$	Stellar scale, fusion-sustained plasma structures, star geometry
T31	31	$\sim 10^{13}$	Interstellar scale, nebular structures, stellar cluster geometry
T37	37	$\sim 10^{20}$	Galactic scale, spiral arm structure, galactic field geometry
T41	41	$\sim 10^{26}$	Observable universe scale, cosmic web nodes, large-scale structure

Section 3: The Five Density Bands

Every structure at every tier occupies one of five density bands, determined by its D_r value. These bands are not arbitrary divisions — each represents a physically distinct substrate stress regime with a characteristic structural behavior and a characteristic relationship to the UBL equilibrium state.

Band Name	D_r Range	Physical Description	Structural Behavior	UBL Proximity
Ghost	$D_r < 0.01$	Near-void. Local density is less than 1% of the tier's	Very slow dissipation. The structure leaks energy gradually	Far below 1.0. Deeply

Band Name	D _r Range	Physical Description	Structural Behavior	UBL Proximity
		critical density. Substrate pressure vastly exceeds local structure density. The structure is a whisper of organized frequency in an ocean of empty substrate — extremely sparse and spread out.	into the substrate but is not actively compressed. Lifetimes are long because there is no aggressive collapse driver — only slow entropy-driven dispersion.	underdense, maximum free fluctuation territory.
Quiet	0.01 to 0.10	Low density, substrate-dominant. Structures have shape but low mass concentration. The structural equivalent of a diffuse gas cloud — organized but sparse.	Moderate dispersion rate. Structures slowly bleed coherence into the substrate but can maintain identity for extended periods.	Below 1.0 but approaching it from below.
Active	0.10 to 10.0	The living zone. This is where most stable physical structures in the universe reside — from atoms to stars to galaxies at their respective tiers. D _r = 1.0 sits at the center of this band, the perfect equilibrium point.	At D _r = 1.0, lifetime is theoretically infinite. Moving away from 1.0 in either direction within this band shortens lifetime, but the Active band still supports the longest-lived normal structures. This is the sweet spot of UST stability.	Straddling the equilibrium line. D _r = 1.0 is the substrate-balanced state.
Hot	10.0 to 1,000.0	High-density compression zone. The structure is significantly overdense relative to its tier equilibrium. Compressed stellar cores, dense molecular clouds, high-	Active compression driving shortened lifetime. The substrate is being squeezed inward. Structures here radiate energy rapidly, change form, or undergo phase transitions. Short lifetimes.	D _r moving upward toward conditions that approach black-hole-

Band Name	D_r Range	Physical Description	Structural Behavior	UBL Proximity
		energy particle collisions at their respective scales.		like internal states.
Critical	D_r > 1,000.0	Extreme compression. More than 1,000 times the equilibrium density at the tier. The pre-collapse zone. Neutron star interiors, cores of black hole formation events, extreme early-universe density peaks.	Extremely short lifetimes. The structure is in free-fall toward collapse. The substrate cannot maintain the frequency lock at this density differential. The structure either explodes (releases energy outward) or collapses to a higher-compression state no longer coherent at this tier.	Approaching UBL = 1. This is the approach to the black hole / bounce threshold.

Section 4: The Lifetime Formula

4.1 Core Formula

This is the central equation of this paper. It takes a prime tier number and a density ratio as input, and returns the predicted lifetime of a coherent structure at that tier in that density environment.

$$L(p, D_r) = \tau_{\text{base}}(p) \times p^{\alpha} / |D_r - 1|^{\beta(\text{band})}$$

Plain English: Lifetime equals the baseline time constant for this tier, multiplied by the tier's prime number raised to the tier-scaling power, divided by the absolute distance of the density ratio from perfect equilibrium raised to the density-sensitivity power for this band.

Symbol Key:

- **$L(p, D_r)$** — Predicted lifetime of the structure, in seconds. This is the decay time constant of the frequency lock at the given tier and density ratio.
- **p** — Prime number of the tier (e.g., $p = 7$ for Tier 7, $p = 41$ for Tier 41).
- **$\tau_{\text{base}}(p)$** — Baseline lifetime constant for prime tier p . This is the characteristic timescale of the tier — derived from the ratio of the tier's characteristic length scale to the speed of light. Anchored physically to the light-crossing time across the tier's fundamental structure.
- **α** — Tier-scaling exponent. In UST geometry, $\alpha = 1.618$ (the golden-ratio coupling constant for substrate resonance amplification across tiers). Reflects the self-similar geometry of the substrate's hierarchical frequency structure.
- **$|D_r - 1|$** — Absolute deviation of the density ratio from equilibrium. Zero only at perfect balance ($D_r = 1.0$), where the formula gives infinite lifetime. As D_r moves away from 1.0 in either direction, this term grows and lifetime shrinks.
- **$\beta(\text{band})$** — Density sensitivity exponent. Varies by density band. Ghost = 0.25, Quiet = 0.50, Active = 1.00, Hot = 1.50, Critical = 2.00. Higher beta means lifetime falls faster as D_r moves from 1.0. The Critical band has the steepest sensitivity — small density increases past $D_r = 1000$ slash lifetime dramatically.

4.2 Why the Golden Ratio for Alpha

In UST, the tier hierarchy is a self-similar resonance structure. Each prime tier is not just a bigger version of the previous — it is a geometrically amplified resonance level. The substrate's frequency architecture follows a phi (golden ratio = 1.618...) scaling because phi is the limit of the ratio of consecutive Fibonacci numbers, and Fibonacci sequences emerge naturally whenever a recursive substrate tries to pack frequency modes as efficiently as possible without destructive interference. Phi is the packing efficiency optimum for recursive wave systems. Alpha = 1.618 is the substrate's own geometry encoded into the lifetime formula — not a parameter tuned to fit data, but the natural geometry of self-similar recursive frequency nesting.

4.3 Why $|D_r - 1|$ and Not D_r Directly

The structure does not feel its absolute density — it feels how far it is from equilibrium. A structure at $D_r = 0.001$ and a structure at $D_r = 1,001$ are both roughly one unit away from equilibrium in absolute magnitude, but their physical stress is fundamentally different — hence the asymmetric beta values. The Ghost band's low beta (0.25) reflects that an underdense structure drifts toward dissolution slowly — there is no active crushing force, only the passive pressure of the surrounding substrate gently pulling the structure apart. The Critical band's high beta (2.00) reflects that an overdense structure is being actively crushed by substrate pressure, and every additional unit of D_r past equilibrium accelerates that crushing by the square — a nonlinear collapse driver that makes the Critical band catastrophically fast compared to the Ghost band.

4.4 The τ_{base} Anchors

For each tier, τ_{base} is derived from the light-crossing time across the tier's characteristic length scale.

$$\tau_{\text{base}}(p) = L_{\text{scale}}(p) / c$$

Plain English: The baseline time constant equals the characteristic length scale of the tier divided by the speed of light.

Symbol Key:

- **$L_{\text{scale}}(p)$** — Characteristic length scale of prime tier p , in meters. This is the physical size of the structures that form at this tier.
- **c** — Speed of light: 3.0×10^8 m/s. Sets the minimum time for coherence information to propagate across the structure.

The light-crossing time is the fundamental minimum time for a signal to traverse the structure — the floor on how quickly information about a density imbalance can propagate across the structure and begin to trigger coherence loss. It is the minimum possible response time of the structure to stress.

Tier	Prime (p)	Characteristic Scale (m)	tau_base (s)	Physical Anchor Meaning
T2	2	1.6×10^{-35}	$\sim 5.0 \times 10^{-44}$	Planck time — the substrate lattice oscillation period
T3	3	1.0×10^{-30}	$\sim 3.0 \times 10^{-39}$	Sub-Planck foam oscillation timescale
T5	5	1.0×10^{-18}	$\sim 3.0 \times 10^{-27}$	Particle formation / quark confinement scale
T7	7	1.0×10^{-15}	$\sim 3.0 \times 10^{-24}$	Nuclear / hadronic interaction time
T11	11	1.0×10^{-10}	$\sim 3.0 \times 10^{-19}$	Atomic electron orbital period scale
T13	13	1.0×10^{-9}	$\sim 3.0 \times 10^{-18}$	Molecular bond vibration period
T17	17	1.0×10^{-6}	$\sim 3.0 \times 10^{-15}$	Cellular process time floor
T19	19	1.0×10^0	$\sim 3.0 \times 10^{-9}$	Organism-scale coherence signal time
T23	23	1.0×10^6	$\sim 3.0 \times 10^{-3}$	Planetary-scale pressure wave crossing time
T29	29	1.0×10^9	$\sim 3.0 \times 10^0$	Stellar photon traversal time
T31	31	1.0×10^{13}	$\sim 3.0 \times 10^4$	Interstellar light-crossing floor (~8 hours)
T37	37	1.0×10^{20}	$\sim 3.0 \times 10^{11}$	Galactic light crossing (~10,000 years)
T41	41	1.0×10^{26}	$\sim 3.0 \times 10^{17}$	Observable universe light crossing (~10 billion years)

Section 5: Density Band Representative D_r Values

For computing all per-tier lifetime results in Section 6, the following fixed representative D_r values are used for each band. These are not averages — they are representative midpoints chosen to characterize typical conditions in each band.

Band	Representative D_r	$ D_r - 1 $	β	$ D_r - 1 ^{\beta}$	Physical Rationale
Ghost	0.001	0.999	0.25	~ 0.9998	Deep void — density 0.1% of equilibrium
Quiet	0.050	0.950	0.50	~ 0.9747	Sparse structure — 5% of equilibrium density
Active	2.000	1.000	1.00	1.0000	One unit above equilibrium — standard operating point
Hot	100.0	99.0	1.50	~ 985.1	100x equilibrium density — compressed zone
Critical	10,000.0	9,999.0	2.00	$\sim 9.998 \times 10^7$	10,000x equilibrium — pre-collapse extreme

The p^{α} multipliers for each tier, using $\alpha = 1.618$, are computed below. These are tier-fixed values used in all subsequent calculations.

Tier	Prime (p)	$p^{1.618}$ (computed)
T2	2	3.07
T3	3	5.84
T5	5	13.0

Tier	Prime (p)	$p^{1.618}$ (computed)
T7	7	22.8
T11	11	44.7
T13	13	55.1
T17	17	78.5
T19	19	90.8
T23	23	116.0
T29	29	154.0
T31	31	166.0
T37	37	205.0
T41	41	232.0

Band ratio note:

Because the band multipliers ($|D_r - 1|^{\beta}$) depend only on D_r and β — not on p or τ_{base} — the ratio between bands is the same for every tier. Ghost is approximately 1.0002 times Active. Quiet is approximately 1.026 times Active. Hot is approximately $1/985$ of Active. Critical is approximately $1/(9.998 \times 10^7)$ of Active. These ratios are tier-universal.

Section 6: Per-Tier Analysis — All 13 Prime Tiers

Tier 2 (T2): Planck Floor

Tier 2 is the bottom of the UST hierarchy — the Planck scale, where the substrate lattice itself is the structure. At 10^{-35} meters, there are no particles here, no fields, no forces as physics currently defines them. There is only the raw oscillation of the substrate medium at its minimum lattice spacing.

Structures at T2 are the briefest organization events possible: micro-collapses, vacuum fluctuations, and density knots in the substrate foam that either immediately dissipate or, when they achieve frequency coherence near $D_r = 1.0$, briefly hold together long enough to seed T3 structures above them. The Ghost band at T2 represents the vast sub-Planck vacuum — near-zero density regions that are effectively pure substrate with no organized energy. The Critical band at T2 represents the extreme density events at the Planck scale — micro black hole nucleation events, quantum gravity domain walls.

$$\tau_{\text{base}} = 5.0 \times 10^{-44} \text{ s} \mid p = 2 \mid p^{\alpha} = 3.07$$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	3.07	$5.0\text{e-}44 \times 3.07 / (0.999)^{0.25} = 1.535\text{e-}43 / 0.9998$	~1.54 x 10 ⁻⁴³ s	Near-Planck floor void fluctuations — essentially same timescale as Active band. Void regions at T2 are not short-lived.
Quiet	0.050	0.950	0.50	3.07	$5.0\text{e-}44 \times 3.07 / (0.950)^{0.50} = 1.535\text{e-}43 / 0.9747$	~1.57 x 10 ⁻⁴³ s	Sparse sub-Planck structures. Slightly longer than Active — low beta means underdensity barely penalizes lifetime.
Active	2.000	1.000	1.00	3.07	$5.0\text{e-}44 \times 3.07 / 1.000 = 1.535\text{e-}43$	~1.54 x 10 ⁻⁴³ s	Planck-scale coherence events at standard operating density. This is the Planck time floor of UST structure.

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Hot	100.0	99.0	1.50	3.07	$1.535e-43 / (99.0)^{1.50} = 1.535e-43 / 985.1$	~1.56 x 10 ⁻⁴⁶ s	Planck-scale high-density compression events. Three orders shorter than active — these are sub-Planck-time collapses.
Critical	10,000	9,999	2.00	3.07	$1.535e-43 / (9999.0)^{2.00} = 1.535e-43 / 9.998e7$	~1.54 x 10 ⁻⁵¹ s	Instantaneous micro-collapses at the substrate lattice — feed the vacuum energy spectrum. Seven orders below Planck time.

At the Planck floor, even Active-band lifetimes are sub-Planck-time in scale — these are the shortest-lived structural events in UST. The Ghost and Active bands converge nearly identically because the deviation from equilibrium at D_r = 0.001 and D_r = 2.0 produces nearly the same |D_r - 1| magnitude. Critical-band lifetimes at T2 are approximately 10⁻⁵¹ seconds — seven orders shorter than Planck time — representing instantaneous substrate micro-collapses that feed the vacuum energy spectrum. There is no slower process in UST than the Ghost-band drift at T2, and no faster collapse than the Critical-band snap at T2.

Tier 3 (T3): Quantum Foam

Tier 3 sits just above the Planck floor at the 10⁻³⁰ meter scale — the quantum foam regime. This is where the substrate's self-organizing turbulence lives. The foam at T3 is not randomness — it is the substrate generating brief density fluctuations that test the stability conditions for T5 particle formation above it. Ghost-band T3 structures are the ultra-dilute substrate fluctuations that form and dissolve continuously in what standard physics calls the quantum vacuum. Active-band T3 structures are the brief coherence events that, when they hit near D_r = 1.0, stabilize long enough to seed more complex

organization. Critical-band T3 events are the extreme compression spikes that generate virtual particle pair production signatures measurable at the quantum level.

$$\tau_{\text{base}} = 3.0 \times 10^{-39} \text{ s} \mid p = 3 \mid p^{\alpha} = 5.84$$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	5.84	$3.0 \times 10^{-39} \times 5.84 / 0.9998 = 1.752 \times 10^{-38} / 0.9998$	$\sim 1.75 \times 10^{-38} \text{ s}$	Ultra-dilute foam fluctuations. Long-lived by T3 standards — slow passive dispersion into substrate background.
Quiet	0.050	0.950	0.50	5.84	$3.0 \times 10^{-39} \times 5.84 / 0.9747 = 1.752 \times 10^{-38} / 0.9747$	$\sim 1.80 \times 10^{-38} \text{ s}$	Sparse foam structures — slightly extended lifetime consistent with low-beta underdensity behavior.
Active	2.000	1.000	1.00	5.84	$3.0 \times 10^{-39} \times 5.84 / 1.000 = 1.752 \times 10^{-38}$	$\sim 1.75 \times 10^{-38} \text{ s}$	Standard quantum foam coherence events — the characteristic T3 timescale. Seeds T5 particle formation at equilibrium.
Hot	100.0	99.0	1.50	5.84	$1.752 \times 10^{-38} / 985.1$	$\sim 1.78 \times 10^{-41} \text{ s}$	Compressed foam events — high-density substrate spikes consistent with virtual particle timescales.
Critical	10,000	9,999	2.00	5.84	$1.752 \times 10^{-38} / 9.998 \times 10^7$	$\sim 1.75 \times 10^{-46} \text{ s}$	Extreme foam compression — instantaneous substrate collapses well below

Band	D_r	D_r - 1	beta	p^alpha	Formula Computation	Lifetime Result	Physical Interpretation
							virtual particle observation windows.

Quantum foam structures at T3 survive slightly longer than T2 — five orders of magnitude in tau_base and a larger prime amplifier push all T3 lifetimes approximately five orders above T2 values. Even so, these are events well below any measurable timescale. The foam at this tier is the substrate's self-organizing turbulence — brief density fluctuations that either dissipate back into the substrate or, when they hit the Active band near D_r = 1.0, stabilize long enough to seed T5 particle formation. The Ghost band at T3 has nearly the same lifetime as the Active band — voids in the quantum foam are just as long-lived as coherent foam structures, because dispersion at this scale is driven by the same passive substrate pressure regardless of which direction D_r deviates from 1.0.

Tier 5 (T5): Particle Formation

Tier 5 is where recognizable physics begins. At the 10^-18 meter scale, this is the domain of fundamental particle formation — quark-level resonance loops, gluon field geometry, and the pre-hadronic substrate organization that produces what standard physics calls elementary particles. Ghost-band T5 structures are free quarks in deconfined quark-gluon plasma — isolated frequency fragments without a stable hadronic container. Active-band T5 structures near D_r = 1.0 are stable bound particles like protons and electrons — the universe's most durable organizational achievements. Hot-band T5 structures are the brief interiors of particle collisions in high-energy accelerators. Critical-band T5 events are the creation and annihilation of virtual particle pairs in extreme fields.

tau_base = 3.0 x 10^-27 s | p = 5 | p^alpha = 13.0

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	13.0	$3.0e-27 \times 13.0 / 0.9998 = 3.90e-26 / 0.9998$	~3.90 x 10 ⁻²⁶ s	Deconfined quark-level fragments drifting in plasma — slow dispersion, long ghost-band survival consistent with quark-gluon plasma persistence.
Quiet	0.050	0.950	0.50	13.0	$3.0e-27 \times 13.0 / 0.9747 = 3.90e-26 / 0.9747$	~4.00 x 10 ⁻²⁶ s	Sparse particle-scale structures — short-lived unstable particles with low mass density relative to their tier equilibrium.
Active	2.000	1.000	1.00	13.0	$3.0e-27 \times 13.0 / 1.000 = 3.90e-26$	~3.90 x 10 ⁻²⁶ s	Baseline T5 particle lifetime at D _r = 2.0. Unstable short-lived particles sit here. Stable particles (protons) achieve D _r near 1.0, giving lifetimes orders of magnitude longer.
Hot	100.0	99.0	1.50	13.0	$3.90e-26 / 985.1$	~3.96 x 10 ⁻²⁹ s	High-energy collision interiors — timescale of particle creation events in accelerator collisions. Matches observed particle interaction timescales.
Critical	10,000	9,999	2.00	13.0	$3.90e-26 / 9.998e7$	~3.90 x 10 ⁻³⁴ s	Instantaneous particle-scale collapse events — pair creation/annihilation timescales under extreme field conditions.

At the particle tier, Active-band structures have lifetimes around 10^{-26} seconds — consistent with the lifetime of very short-lived unstable particles in standard physics. Stable particles like protons achieve D_r very close to 1.0 — essentially living on the knife-edge of perfect equilibrium, which is why their lifetimes are astronomically long in practice (proton lifetime greater than 10^{34} years as measured experimentally). The Critical band at T5 yields 10^{-34} second lifetimes — instantaneous by any measure, which is the timescale of particle creation-annihilation events in extreme density conditions. The Ghost band at T5, representing deconfined quarks in a plasma environment, survives as long as the Active band — consistent with the observed persistence of quark-gluon plasma in heavy-ion collision experiments before hadronization.

Tier 7 (T7): Nuclear Scale

Tier 7 at 10^{-15} meters is the nuclear domain — protons, neutrons, and their interactions within atomic nuclei. Structures at this tier are hadronic: they are the composite frequency locks formed when T5 particle resonances bind together at the nuclear scale. Ghost-band T7 structures are free nucleons — protons or neutrons ejected from nuclei in energetic reactions, no longer bound to nuclear equilibrium. Active-band T7 structures are stable nuclei sitting close to $D_r = 1.0$ — the most stable isotopes of every element. Hot-band T7 structures are the compressed nuclear matter inside neutron stars or the brief nuclear fireball of a heavy-ion collision. Critical-band T7 events are the microseconds inside a core-collapse supernova before neutron star formation.

$\tau_{base} = 3.0 \times 10^{-24} \text{ s} \mid p = 7 \mid p^{\alpha} = 22.8$

Band	D_r Rep.	$ D_r - 1 $	beta	p^{α}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	22.8	$3.0e-24 \times 22.8 / 0.9998 = 6.84e-23 / 0.9998$	$\sim 6.84 \times 10^{-23} \text{ s}$	Free nucleons in low-density environments. Long-lived ghost survival consistent with free neutron's measured 14-minute beta-

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
							decay lifetime being a higher-tier phenomenon.
Quiet	0.050	0.950	0.50	22.8	$3.0e-24 \times 22.8 / 0.9747 = 6.84e-23 / 0.9747$	$\sim 7.02 \times 10^{-23}$ s	Sparse nuclear-scale structures — loosely bound nuclear clusters at the edge of stability (exotic nuclei, halo nuclei).
Active	2.000	1.000	1.00	22.8	$3.0e-24 \times 22.8 / 1.000 = 6.84e-23$	$\sim 6.84 \times 10^{-23}$ s	Nuclear resonance baseline at D _r = 2.0. Actual stable nuclei sit at D _r near 1.0, giving lifetimes far exceeding this baseline.
Hot	100.0	99.0	1.50	22.8	$6.84e-23 / 985.1$	$\sim 6.94 \times 10^{-26}$ s	Dense nuclear matter in neutron star mergers or heavy-ion collisions — brief nuclear fireball before reorganizing to a new phase.
Critical	10,000	9,999	2.00	22.8	$6.84e-23 / 9.998e7$	$\sim 6.84 \times 10^{-31}$ s	Instantaneous nuclear-scale compression events — the timescale of nuclear collapse during core-collapse supernova dynamics.

Nuclear structures at the Active band survive approximately 7×10^{-23} seconds at the baseline D_r = 2.0 operating point. Actual nuclei in stable isotopes sit at D_r very close to 1.0, explaining their essentially infinite stability on human timescales. Hot-band nuclei — extreme compression events like those in neutron star mergers — drop to approximately 7×10^{-26} second lifetimes, consistent with the brief existence of hot nuclear matter before it reorganizes into a new structural phase. The Critical band at T7, at 6.84×10^{-31} seconds, represents the duration of the nuclear collapse snap itself — the moment

the core of a dying star crosses from nuclear matter into something the substrate can no longer hold as a T7 coherent structure.

Tier 11 (T11): Atomic Scale

Tier 11 at 10^-10 meters is the atomic domain — the scale of electron shell resonance and chemical identity. Every element on the periodic table is a distinct T11 frequency lock: a specific arrangement of electron resonance loops around a nuclear core, stabilized when D_r sits near 1.0 for that element's electron shell geometry. Ghost-band T11 structures are ionized atoms stripped of electrons — just a nucleus in a near-void environment with no electron shell frequency lock. Active-band T11 structures are neutral atoms in stable configurations. Hot-band T11 structures are highly excited atoms near ionization thresholds, or atoms in extreme plasma conditions. Critical-band T11 events are the attosecond dynamics of complete electron shell disruption.

$\tau_{base} = 3.0 \times 10^{-19} \text{ s} \mid p = 11 \mid p^{\alpha} = 44.7$

Band	D_r Rep.	D_r - 1	beta	p^alpha	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	44.7	$3.0e-19 \times 44.7 / 0.9998 = 1.341e-17 / 0.9998$	$\sim 1.34 \times 10^{-17} \text{ s}$	Ionized atoms in ultra-low-density plasma — electron-stripped nuclei. Extended coherence of the nuclear charge pattern even without shell lock.
Quiet	0.050	0.950	0.50	44.7	$3.0e-19 \times 44.7 / 0.9747 = 1.341e-17 / 0.9747$	$\sim 1.38 \times 10^{-17} \text{ s}$	Partially ionized atoms in sparse plasma — weakly bound outer electron shells with slow coherence loss.

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Active	2.000	1.000	1.00	44.7	$3.0e-19 \times 44.7 / 1.000 = 1.341e-17$	$\sim 1.34 \times 10^{-17}$ s (~13 attoseconds)	Atomic electron dynamics baseline. The attosecond timescale directly measured in ultrafast electron physics experiments.
Hot	100.0	99.0	1.50	44.7	$1.341e-17 / 985.1$	$\sim 1.36 \times 10^{-20}$ s	Highly compressed atomic structures in dense plasma — electron shell disruption timescale in intense laser fields.
Critical	10,000	9,999	2.00	44.7	$1.341e-17 / 9.998e7$	$\sim 1.34 \times 10^{-25}$ s	Instantaneous complete atomic collapse — total electron shell destruction under extreme compression.

At the atomic tier, Active-band lifetime is approximately 13 attoseconds — in the same range as attosecond electron dynamics measured in modern ultrafast physics experiments. This is not a coincidence in UST. Attosecond physics is detecting the characteristic lifetime of T11 Active-band fluctuations — the substrate's own resonance timescale at the atomic scale. Ionized atoms pushed into the Quiet or Ghost band by charge stripping survive longer by this formula, consistent with the observed extended lifetime of excited ionic states in low-density plasmas. The Hot band at T11, at approximately 10^{-20} seconds, represents the timescale of electron shell disruption in intense focused laser pulses — matching the picosecond-to-femtosecond windows observed in strong-field atomic physics.

Tier 13 (T13): Molecular Scale

Tier 13 at 10^{-9} meters is the molecular domain — chemical bonds, nanoscale structures, and the emergence of molecular identity from atomic combination. T13 is the tier at which chemistry happens. Ghost-band T13 structures are isolated molecular fragments in near-vacuum conditions — radicals, dissociated molecular pieces, or sparse gas-phase molecules in interstellar clouds. Active-band T13 structures are stable molecules in typical chemical environments. Hot-band T13 structures are molecules under intense pressure or in high-energy collision events — reactive intermediates in combustion, photodissociation products, or shock-compressed molecular matter. Critical-band T13 events are femtosecond bond-breaking dynamics.

$$\tau_{\text{base}} = 3.0 \times 10^{-18} \text{ s} \mid p = 13 \mid p^{\alpha} = 55.1$$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	55.1	$3.0\text{e-}18 \times 55.1 / 0.9998 = 1.653\text{e-}16 / 0.9998$	$\sim 1.653 \times 10^{-16} \text{ s}$ (~165 attoseconds)	Sparse molecular fragments in interstellar conditions — slow drift toward dispersion, long ghost-band survival consistent with molecular cloud persistence.
Quiet	0.050	0.950	0.50	55.1	$3.0\text{e-}18 \times 55.1 / 0.9747 = 1.653\text{e-}16 / 0.9747$	$\sim 1.697 \times 10^{-16} \text{ s}$	Dilute molecular gas — slightly longer than Active, consistent with isolated gas-phase molecule coherence at low density.
Active	2.000	1.000	1.00	55.1	$3.0\text{e-}18 \times 55.1 / 1.000 = 1.653\text{e-}16$	$\sim 1.653 \times 10^{-16} \text{ s}$ (~165 attoseconds)	Molecular resonance baseline. Matches the timescale of femtosecond chemical reaction dynamics observed in

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
							pump-probe spectroscopy.
Hot	100.0	99.0	1.50	55.1	$1.653e-16 / 985.1$	$\sim 1.678 \times 10^{-19} \text{ s}$ (~ 0.17 attoseconds)	Molecules under extreme pressure or intense laser pulses — bond-breaking events in shock-compressed or photodissociated matter.
Critical	10,000	9,999	2.00	55.1	$1.653e-16 / 9.998e7$	$\sim 1.653 \times 10^{-24} \text{ s}$	Instantaneous molecular collapse — total bond destruction under extreme plasma or shock conditions. Sub-attosecond scale.

Molecular resonance structures at T13 have baseline Active-band lifetimes around 165 attoseconds — consistent with the timescale of femtosecond chemical reaction dynamics observable in pump-probe laser spectroscopy. Hot-band molecular structures — those under extreme pressure or in high-energy collision events — decay in under a femtosecond, matching observations of bond-breaking under intense laser pulses or in extreme plasma chemistry environments. The Critical band at T13, at 10^{-24} seconds, is the timescale of total molecular destruction in the most extreme compression scenarios — far below any observable molecular chemistry, representing the instant a molecular frequency lock simply ceases to exist.

Tier 17 (T17): Cellular Scale

Tier 17 at 10^{-6} meters is the cellular and microbial domain — the scale of organelle geometry, cell membrane dynamics, and microbial structural coherence. This is the first tier where what physics calls

biology begins to emerge. Ghost-band T17 structures are isolated membrane fragments, cytoskeletal remnants, or organelle pieces in near-vacuum or extreme dilution conditions. Active-band T17 structures are living cells with intact membrane integrity and organelle organization. Hot-band T17 structures are cells under extreme osmotic pressure, thermal shock, or high-density compression events. Critical-band T17 events are the rapid lysis dynamics of cells under catastrophic mechanical or osmotic failure.

tau_base = 3.0 x 10⁻¹⁵ s | p = 17 | p^{alpha} = 78.5

Band	D_r Rep.	D_r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	78.5	3.0e-15 x 78.5 / 0.9998 = 2.355e-13 / 0.9998	~2.355 x 10 ⁻¹³ s (~235 picoseconds)	Isolated membrane vesicles in ultra-low-density suspension — slow passive dispersion matching observed persistence of cell membrane vesicles in dilute conditions.
Quiet	0.050	0.950	0.50	78.5	3.0e-15 x 78.5 / 0.9747 = 2.355e-13 / 0.9747	~2.416 x 10 ⁻¹³ s (~242 picoseconds)	Dilute cytoskeletal fragments — slightly extended lifetime relative to Active, consistent with low-density beta suppression.
Active	2.000	1.000	1.00	78.5	3.0e-15 x 78.5 / 1.000 = 2.355e-13	~2.355 x 10 ⁻¹³ s (~235 picoseconds)	Cellular baseline coherence timescale. Protein folding events and membrane oscillation modes fall in this range — consistent

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
							with molecular dynamics simulations of cellular processes.
Hot	100.0	99.0	1.50	78.5	$2.355e-13 / 985.1$	$\sim 2.391 \times 10^{-16} \text{ s}$ (~ 0.24 femtoseconds)	Cells under extreme compression or thermal shock — rapid structural disruption timescale matching intense laser ablation of biological tissue.
Critical	10,000	9,999	2.00	78.5	$2.355e-13 / 9.998e7$	$\sim 2.355 \times 10^{-21} \text{ s}$	Instantaneous cellular collapse — catastrophic lysis or extreme shock wave impact at the cellular scale. Far below measurable biological dynamics.

Cellular-scale structures at T17 operate on picosecond baseline timescales. Individual protein folding events and membrane oscillation modes fall in this range — consistent with molecular dynamics simulations showing protein conformational changes happening on the 100-picosecond to nanosecond scale. The Active-band baseline of 235 picoseconds is the T17 substrate's own resonance clock — the interval at which coherence information traverses a cellular-scale structure. Ghost-band cellular structures — isolated cell membrane fragments in near-vacuum conditions or dilute cytoskeletal fragments — survive for extended periods as predicted, matching the observed persistence of cell membrane vesicles in ultra-low-density suspension conditions where there is no external pressure forcing rapid dispersion.

Tier 19 (T19): Organism Scale

Tier 19 at the 10^0 meter scale is the organism domain — the scale of integrated living systems, macro-biological coherence, and the largest single-body frequency locks in nature. A human body, a whale, a tree — these are all T19 Active-band structures, organized frequency patterns that persist by actively maintaining their density ratio near equilibrium through metabolic processes. Ghost-band T19 structures are diffuse biological matter in extreme dilution — single cells or small clusters in open ocean or atmospheric conditions. Hot-band T19 structures are organisms under extreme pressure or gravitational compression. Critical-band T19 events are catastrophic organism-scale compression or disintegration dynamics.

$$\tau_{\text{base}} = 3.0 \times 10^{-9} \text{ s} \mid p = 19 \mid p^{\alpha} = 90.8$$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	90.8	$3.0 \times 10^{-9} \times 90.8 / 0.9998 = 2.724 \times 10^{-7} / 0.9998$	$\sim 2.724 \times 10^{-7} \text{ s}$ (~272 nanoseconds)	Diffuse organism-scale matter in extreme dilution — the decay constant of an organism-tier resonance in near-void conditions.
Quiet	0.050	0.950	0.50	90.8	$3.0 \times 10^{-9} \times 90.8 / 0.9747 = 2.724 \times 10^{-7} / 0.9747$	$\sim 2.793 \times 10^{-7} \text{ s}$ (~279 nanoseconds)	Sparse organism-scale structures — slightly extended lifetime at the sparse end of the living zone.
Active	2.000	1.000	1.00	90.8	$3.0 \times 10^{-9} \times 90.8 / 1.000 = 2.724 \times 10^{-7}$	$\sim 2.724 \times 10^{-7} \text{ s}$ (~272 nanoseconds)	Baseline T19 resonance decay constant at D _r = 2.0. This is not biological lifespan — it is the decay

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
							time of an unregulated organism-tier frequency lock at this density offset.
Hot	100.0	99.0	1.50	90.8	$2.724e-7 / 985.1$	$\sim 2.766 \times 10^{-10} \text{ s}$ (~ 0.277 nanoseconds)	Organism-scale structure under extreme compression — catastrophic mechanical impact dynamics at the macroscale.
Critical	10,000	9,999	2.00	90.8	$2.724e-7 / 9.998e7$	$\sim 2.724 \times 10^{-15} \text{ s}$ (~ 2.7 femtoseconds)	Instantaneous organism-scale collapse under extreme compression — outside any biological observation window.

Critical note on T19 lifetimes:

The nanosecond timescales above are the baseline decay constants of the T19 resonance pattern at $D_r = 2.0$ — not an organism's biological lifespan. Real organisms persist for years to centuries by continuously maintaining D_r near 1.0 through metabolic regulation. Metabolism is the organism's mechanism for actively pumping energy into the resonance to hold the density ratio at equilibrium against the substrate's tendency to disperse the structure. The formula gives the decay time constant of an unregulated organism-scale frequency lock displaced from equilibrium by a factor of 2. Lose the metabolic regulation — stop eating, stop breathing, stop maintaining internal pressure gradients — and the T19 frequency lock begins decaying on a timescale that the formula predicts. The biological

lifespan is not $L(19, 2.0)$. It is the integral over a lifetime of continuous D_r maintenance near 1.0, which can extend for decades.

Tier 23 (T23): Planetary Scale

Tier 23 at 10^6 meters is the planetary domain — the scale of crustal dynamics, atmospheric coherence, and planetary-scale pressure structures. At this tier, the structures are not solid objects but coherent density patterns: standing pressure waves in planetary interiors, atmospheric resonance modes, tectonic stress coherence regions. Ghost-band T23 structures are proto-planetary gas clouds in near-void conditions — the earliest stages of planetary material aggregation in a molecular cloud. Active-band T23 structures are mature planets with stable internal pressure gradients. Hot-band T23 structures are planetary cores under extreme compression or shock-loaded regions during large impact events. Critical-band T23 events are the rapid dynamics of planetary disruption scenarios.

$\tau_{\text{base}} = 3.0 \times 10^{-3} \text{ s} \mid p = 23 \mid p^\alpha = 116.0$

Band	D_r Rep.	$ D_r - 1 $	beta	p^α	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	116.0	$3.0e-3 \times 116.0 / 0.9998 = 0.348 / 0.9998$	$\sim 0.348 \text{ s}$	Diffuse proto-planetary cloud density structures — sub-second decay constant of a near-void planetary-scale resonance mode.
Quiet	0.050	0.950	0.50	116.0	$3.0e-3 \times 116.0 / 0.9747 = 0.348 / 0.9747$	$\sim 0.357 \text{ s}$	Sparse planetary-scale density structures — molecular cloud cores transitioning toward

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
							protoplanetary disk organization.
Active	2.000	1.000	1.00	116.0	$3.0e-3 \times 116.0 / 1.000 = 0.348$	~0.348 s (~348 milliseconds)	Planetary-scale pressure pulse coherence — shockwaves, seismic standing wave modes that form and dissipate on sub-second timescales.
Hot	100.0	99.0	1.50	116.0	$0.348 / 985.1$	~3.53 x 10 ⁻⁴ s (~0.35 milliseconds)	Planetary core under extreme compression — shock pulse coherence in high-pressure impact dynamics. Millisecond-scale shockwave fronts in large impact events.
Critical	10,000	9,999	2.00	116.0	$0.348 / 9.998e7$	~3.48 x 10 ⁻⁹ s (~3.5 nanoseconds)	Instantaneous planetary-scale collapse — extreme compression event that disrupts the planetary-tier resonance entirely in nanoseconds.

Planetary-scale Active-band resonance structures have decay constants of about 348 milliseconds at the baseline D_r = 2.0 offset. This is the timescale of planetary-scale pressure pulse coherence — shockwaves, seismic waves, and atmospheric standing wave modes that form and dissipate on sub-second timescales before breaking into turbulent dispersion. Ghost-band planetary structures — giant diffuse proto-planetary gas clouds in near-void conditions — have comparable sub-second timescales at the T23 resonance level, meaning they are slow to disperse at the substrate level. The Critical band at

T23, yielding 3.5 nanoseconds, represents the maximum coherence time of an extreme planetary-scale compression event — the instant of maximum compression during a hypervelocity planetary impact before the energy releases outward in an explosion.

Tier 29 (T29): Stellar Scale

Tier 29 at 10^9 meters is the stellar domain — fusion-sustained plasma structures, photospheric dynamics, and the large-scale geometry of star-scale resonance. The sun at this tier is an Active-band structure sitting near D_r = 1.0, held there by the continuous balance of gravitational compression and fusion-driven pressure. Ghost-band T29 structures are proto-stellar gas filaments in molecular clouds — low-density stellar-scale matter before gravitational collapse achieves critical density. Active-band T29 structures are main-sequence stars with stable fusion cores. Hot-band T29 structures are stellar cores during red-giant expansion or pre-supernova compression phases. Critical-band T29 events are the final seconds of core collapse in a massive star.

$\tau_{base} = 3.0 \times 10^0 \text{ s} \mid p = 29 \mid p^{\alpha} = 154.0$

Band	D_r Rep.	D_r - 1	beta	p^alpha	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	154.0	$3.0 \times 154.0 / 0.9998 = 462.0 / 0.9998$	~462.1 s (~7.7 minutes)	Diffuse proto-stellar filaments in molecular cloud environments — the decay constant of a ghost-band stellar-scale resonance mode.
Quiet	0.050	0.950	0.50	154.0	$3.0 \times 154.0 / 0.9747 = 462.0 / 0.9747$	~474.0 s (~7.9 minutes)	Sparse stellar-scale structures — protostellar cores at the low end of active accretion.

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Active	2.000	1.000	1.00	154.0	$3.0 \times 154.0 / 1.000 = 462.0$	~462 s (~7.7 minutes)	Stellar resonance baseline. Matches the solar p-mode oscillation period (3 to 20 minutes). UST reads these as the T29 Active-band lifetime signature.
Hot	100.0	99.0	1.50	154.0	$462.0 / 985.1$	~0.469 s (~0.47 seconds)	Pre-supernova core compression phases — sub-second coherence matching observed millisecond-to-second timescales of core-collapse supernova dynamics.
Critical	10,000	9,999	2.00	154.0	$462.0 / 9.998e7$	~4.62 x 10 ⁻⁶ s (~4.6 microseconds)	Stellar core in free-fall collapse — the final microseconds of a T29 frequency lock before neutron star formation or black hole formation.

Stellar Active-band resonance baseline is approximately 7.7 minutes — the characteristic timescale for coherent density perturbations to traverse and dissipate across a stellar-scale structure. This is directly consistent with the solar p-mode oscillation period — the Sun's acoustic resonance modes have periods of 3 to 20 minutes. UST interprets these oscillations as the substrate's T29 Active-band lifetime signature: the Sun is continuously cycling through small D_r perturbations at this timescale, maintaining its overall frequency lock near D_r = 1.0 while individual density modes form and dissolve with a characteristic 7.7-minute decay constant. Hot-band stellar structures — stellar cores at extreme compression during pre-supernova phases — have sub-second coherence times, matching the observed

millisecond-to-second timescale of core-collapse supernova dynamics as measured by neutrino burst detectors.

Tier 31 (T31): Interstellar Scale

Tier 31 at 10^{13} meters is the interstellar domain — the scale of nebular structures, HII regions, supernova-driven shockwaves, and stellar cluster geometry. Ghost-band T31 structures are the ultra-low-density cosmic web bridges between galactic clusters — the most tenuous organized matter in the universe. Active-band T31 structures are coherent interstellar medium density waves and nebular filaments. Hot-band T31 structures are compressed interstellar medium regions behind supernova shockwaves or at cloud-cloud collision boundaries. Critical-band T31 events are the brief extreme-compression dynamics inside supernova remnant blast cores.

$$\tau_{\text{base}} = 3.0 \times 10^4 \text{ s} \mid p = 31 \mid p^{\alpha} = 166.0$$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	166.0	$3.0e4 \times 166.0 / 0.9998 = 4.98e6 / 0.9998$	$\sim 4.980 \times 10^6 \text{ s}$ (~57.6 days)	Ultra-low-density cosmic web filaments — slow ghost-band drift over weeks before the substrate pressure wins.
Quiet	0.050	0.950	0.50	166.0	$3.0e4 \times 166.0 / 0.9747 = 4.98e6 / 0.9747$	$\sim 5.109 \times 10^6 \text{ s}$ (~59.1 days)	Sparse interstellar medium structures — diffuse gas regions with slow coherence loss over similar timescales.
Active	2.000	1.000	1.00	166.0	$3.0e4 \times 166.0 /$	$\sim 4.98 \times 10^6 \text{ s}$ (~57.6 days)	Interstellar shockwave coherence — supernova-driven shocks maintain

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
					$1.000 = 4.98e6$		structural identity at this scale for ~57 days before breaking into turbulence.
Hot	100.0	99.0	1.50	166.0	$4.98e6 / 985.1$	~5,055 s (~84 minutes)	Dense interstellar medium compression from galactic arm passage or cloud-cloud collision — 84-minute rapid turbulent dissipation consistent with HII region boundary dynamics.
Critical	10,000	9,999	2.00	166.0	$4.98e6 / 9.998e7$	~0.0498 s (~50 milliseconds)	Extreme interstellar compression peak — supernova blast core interior, 50-millisecond coherence window before the blast structure dissolves.

Interstellar structures at T31 have Active-band coherence times of approximately 57 days. This is the timescale of interstellar shockwave coherence — supernova-driven shockwaves maintain their structural identity at the interstellar scale for periods on this order before breaking into turbulence and becoming indistinguishable from the general interstellar medium. Ghost-band interstellar filaments — the ultra-low-density cosmic web bridges between galactic clusters — have comparable 57-day timescales at the T31 resonance level, confirming that the most tenuous structures in the universe are not fragile at their own tier. Hot-band events at T31 — dense interstellar medium compression from galactic arm passage or cloud-cloud collisions — dissipate in 84 minutes, consistent with the rapid turbulent dissipation observed in compressed HII region boundaries where high-density gas is rapidly mixed back into the ambient medium.

Tier 37 (T37): Galactic Scale

Tier 37 at 10^{20} meters is the galactic domain — the scale of spiral arm structure, galactic bar instabilities, tidal streamers, and the large-scale field geometry of entire galaxies. Ghost-band T37 structures are ultra-diffuse galaxies in cosmic void regions — galaxies so dispersed they barely exceed the critical density threshold. Active-band T37 structures are normal spiral and elliptical galaxies with coherent density wave patterns. Hot-band T37 structures are galaxies undergoing starburst events, mergers at extreme tidal compression, or AGN-driven density enhancement. Critical-band T37 events are the brief extreme-compression dynamics of galactic core starburst peaks and gamma-ray burst progenitor events at the galactic scale.

$$\tau_{\text{base}} = 3.0 \times 10^{11} \text{ s} \mid p = 37 \mid p^{\alpha} = 205.0$$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	205.0	$3.0e11 \times 205.0 / 0.9998 = 6.15e13 / 0.9998$	$\sim 6.150 \times 10^{13} \text{ s}$ (~1,950 years)	Ultra-diffuse galaxy coherence in cosmic voids — slow ghost-band dispersion over ~2,000 years at the T37 resonance level.
Quiet	0.050	0.950	0.50	205.0	$3.0e11 \times 205.0 / 0.9747 = 6.15e13 / 0.9747$	$\sim 6.310 \times 10^{13} \text{ s}$ (~2,000 years)	Low-density galactic structures — dwarf galaxies, tidal streams, sparse galactic outskirts with slow coherence bleeding.
Active	2.000	1.000	1.00	205.0	$3.0e11 \times 205.0 /$	$\sim 6.15 \times 10^{13} \text{ s}$	Galactic density wave coherence — spiral arm

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
					$1.000 = 6.15e13$	(~1,950 years)	patterns, bar instabilities, and tidal streamers form, evolve, and dissolve on this timescale at the T37 resonance level.
Hot	100.0	99.0	1.50	205.0	$6.15e13 / 985.1$	~6.24 x 10 ¹⁰ s (~198 years)	Starburst galaxy compression events — galactic-scale density enhancement coherence windows of ~200 years at the structural resonance level.
Critical	10,000	9,999	2.00	205.0	$6.15e13 / 9.998e7$	~6.15 x 10 ⁵ s (~7.1 days)	Extreme galactic-scale compression peak — galactic core starburst or gamma-ray burst progenitor event dynamics. Individual T37 resonance pulse collapses in ~7 days.

Galactic-scale Active-band structures have coherence times of roughly 1,950 years. This is the timescale of large-scale galactic density wave coherence — spiral arm patterns, bar instabilities, and tidal streamer structures that form, evolve, and dissolve on timescales of thousands of years at the structural resonance level. Ghost-band galactic structures — ultra-diffuse galaxies in cosmic void regions — show similarly extended but slow-dissipating coherence over the same 1,950-year timescale, confirming that void-dwelling galaxies are not fragile despite their near-critical underdensity. The Critical-band result of 7.1 days is striking: galactic core starburst episodes can persist for millions of years at the full galactic scale, but the individual T37 resonance pulse within such an event — the coherent compression wave that drives the starburst — collapses in days. This is consistent with observed gamma-ray burst

afterglow structures, where galactic-scale compression peaks produce brief but catastrophically energetic release events.

Tier 41 (T41): Observable Universe Scale

Tier 41 at 10²⁶ meters is the highest tier in UST — the observable universe itself. At this scale, the structures are the cosmic web nodes, large-scale filaments, and the overall density geometry of the universe as a coherent frequency lock on the substrate. Ghost-band T41 structures are the great cosmic voids — vast regions of the universe where matter density falls far below the critical density. Active-band T41 is where the universe as a whole currently resides — slowly expanding, sitting at or near its equilibrium density ratio at this tier. Hot-band T41 events are hypothetical recollapse scenarios where the overall cosmic density rises significantly above equilibrium. Critical-band T41 is the Big Crunch / pre-bounce condition — the universe collapsing back toward UBL = 1 at extreme density.

$\tau_{\text{base}} = 3.0 \times 10^{17} \text{ s} \mid p = 41 \mid p^{\alpha} = 232.0$

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
Ghost	0.001	0.999	0.25	232.0	$3.0e17 \times 232.0 / 0.9998 = 6.96e19 / 0.9998$	~6.960 x 10 ¹⁹ s (~2.21 trillion years)	Great cosmic voids — the most persistent large-scale structures in the universe. Consistent with observations that cosmic voids are among the longest-lived density structures.
Quiet	0.050	0.950	0.50	232.0	$3.0e17 \times 232.0 / 0.9747 = 6.96e19 / 0.9747$	~7.138 x 10 ¹⁹ s (~2.26	Sparse large-scale cosmic filaments — slightly extended lifetime consistent with the

Band	D _r Rep.	D _r - 1	beta	p ^{alpha}	Formula Computation	Lifetime Result	Physical Interpretation
						trillion years)	persistence of low-density filamentary structures in the cosmic web.
Active	2.000	1.000	1.00	232.0	$3.0e17 \times 232.0 / 1.000 = 6.96e19$	~6.96 x 10 ¹⁹ s (~2.2 trillion years)	Universe at D _r = 2.0 — Active-band coherence time. The current universe at ~13.8 billion years is at about 0.6% of this lifetime, a young universe by T41 standards.
Hot	100.0	99.0	1.50	232.0	$6.96e19 / 985.1$	~7.07 x 10 ¹⁶ s (~2.24 billion years)	Hypothetical cosmic recollapse — if the universe's overall D _r reaches 100x equilibrium, the T41 coherence window drops to ~2.24 billion years before full structural reorganization.
Critical	10,000	9,999	2.00	232.0	$6.96e19 / 9.998e7$	~6.96 x 10 ¹¹ s (~22 years)	Big Crunch scenario — if the universe hits D _r = 10,000, the entire T41 resonance collapses in 22 years. The bounce that resets the cycle. No sub-structure survives.

At the observable universe scale — the highest tier in UST — Active-band coherence time is approximately 2.2 trillion years. The current age of the universe at approximately 13.8 billion years means the universe is at roughly 0.6% of its T41 Active-band lifetime if it sits near D_r = 2.0. A universe

sitting closer to $D_r = 1.0$ would have a coherence lifetime that approaches infinity — which is the UST interpretation of cosmological observations showing the universe is in a relatively low-density, slowly-expanding state that could persist for trillions of years. Ghost-band universe structures — vast cosmic voids in the large-scale structure — have similar 2+ trillion year lifetimes, consistent with observations that cosmic voids are among the most persistent large-scale structures in the observable universe.

The Critical-band result of 22 years is the most consequential number in this paper. If the observable universe's overall density ratio ever hit $D_r = 10,000$ — an extreme recollapse scenario in which all matter converges back toward a singularity — the entire T41 resonance structure would collapse in 22 years. This is the Big Crunch / bounce that resets the cycle. The brevity of that 22-year window is itself the mechanism of the bounce: the collapse is too fast for any sub-structure to survive, which is why no information, no pattern, no residual frequency lock persists across cycles. The substrate resets to UBL approaching 0, and the next expansion begins.

Section 7: Cross-Tier Lifetime Comparison Tables

7.1 Master Table — Active Band ($D_r = 2.0$) Lifetimes Across All Tiers

This table shows the Active-band lifetime for all 13 tiers in order. The span from T2 to T41 covers 62 orders of magnitude in predicted structure lifetime.

Tier	Prime (p)	tau_base (s)	p^alpha	Active-Band Lifetime L (s)	Converted Units	Physical Domain
T2	2	5.0×10^{-44}	3.07	1.54×10^{-43}	0.154 yoctoseconds	Planck floor / vacuum lattice
T3	3	3.0×10^{-39}	5.84	1.75×10^{-38}	17.5 yoctoseconds	Quantum foam
T5	5	3.0×10^{-27}	13.0	3.90×10^{-26}	39.0 yoctoseconds	Fundamental particles
T7	7	3.0×10^{-24}	22.8	6.84×10^{-23}	68.4 zeptoseconds	Nuclear / hadronic

Tier	Prime (p)	tau_base (s)	p^alpha	Active-Band Lifetime L (s)	Converted Units	Physical Domain
T11	11	3.0×10^{-19}	44.7	1.34×10^{-17}	13.4 attoseconds	Atomic / electron shells
T13	13	3.0×10^{-18}	55.1	1.65×10^{-16}	165 attoseconds	Molecular / chemical bonds
T17	17	3.0×10^{-15}	78.5	2.36×10^{-13}	236 picoseconds	Cellular / microbial
T19	19	3.0×10^{-9}	90.8	2.72×10^{-7}	272 nanoseconds	Organism scale
T23	23	3.0×10^{-3}	116.0	3.48×10^{-1}	348 milliseconds	Planetary scale
T29	29	3.0×10^0	154.0	4.62×10^2	7.7 minutes	Stellar scale
T31	31	3.0×10^4	166.0	4.98×10^6	57.6 days	Interstellar scale
T37	37	3.0×10^{11}	205.0	6.15×10^{13}	~1,950 years	Galactic scale
T41	41	3.0×10^{17}	232.0	6.96×10^{19}	~2.2 trillion years	Observable universe

7.2 Full Master Table — All Five Bands Across All Tiers (seconds, scientific notation)

Tier	Ghost L (s)	Quiet L (s)	Active L (s)	Hot L (s)	Critical L (s)
T2	1.54×10^{-43}	1.57×10^{-43}	1.54×10^{-43}	1.56×10^{-46}	1.54×10^{-51}
T3	1.75×10^{-38}	1.80×10^{-38}	1.75×10^{-38}	1.78×10^{-41}	1.75×10^{-46}
T5	3.90×10^{-26}	4.00×10^{-26}	3.90×10^{-26}	3.96×10^{-29}	3.90×10^{-34}
T7	6.84×10^{-23}	7.02×10^{-23}	6.84×10^{-23}	6.94×10^{-26}	6.84×10^{-31}
T11	1.34×10^{-17}	1.38×10^{-17}	1.34×10^{-17}	1.36×10^{-20}	1.34×10^{-25}

Tier	Ghost L (s)	Quiet L (s)	Active L (s)	Hot L (s)	Critical L (s)
T13	1.65×10^{-16}	1.70×10^{-16}	1.65×10^{-16}	1.68×10^{-19}	1.65×10^{-24}
T17	2.36×10^{-13}	2.42×10^{-13}	2.36×10^{-13}	2.39×10^{-16}	2.36×10^{-21}
T19	2.72×10^{-7}	2.79×10^{-7}	2.72×10^{-7}	2.77×10^{-10}	2.72×10^{-15}
T23	3.48×10^{-1}	3.57×10^{-1}	3.48×10^{-1}	3.53×10^{-4}	3.48×10^{-9}
T29	4.62×10^2	4.74×10^2	4.62×10^2	4.69×10^{-1}	4.62×10^{-6}
T31	4.98×10^6	5.11×10^6	4.98×10^6	5.06×10^3	4.98×10^{-2}
T37	6.15×10^{13}	6.31×10^{13}	6.15×10^{13}	6.24×10^{10}	6.15×10^5
T41	6.96×10^{19}	7.14×10^{19}	6.96×10^{19}	7.07×10^{16}	6.96×10^{11}

Note:

Ghost, Quiet, and Active band lifetimes are nearly identical at every tier because their $|D_r - 1|$ values (0.999, 0.950, and 1.000 respectively) are all close to 1.0, and their beta exponents (0.25, 0.50, 1.00) are low enough that the denominator stays close to 1. The dramatic drops appear only in the Hot and Critical bands, where both $|D_r - 1|$ is large and beta amplifies the penalty.

7.3 Band-Ratio Table — Lifetime Relative to Active Band

These ratios are tier-independent. They depend only on D_r and beta, not on p or τ_{base} . Every tier in UST has the same band-to-band ratio structure.

Band	$ D_r - 1 ^\beta$	Ratio to Active	Orders of Magnitude	Interpretation
Ghost	~ 0.9998	$\sim 1.0002 \times$ Active	+0.0001	Nearly identical to Active. Voids are as long-lived as equilibrium structures.
Quiet	~ 0.9747	$\sim 1.026 \times$ Active	+0.01	Slightly longer than Active. Low-density structures have marginally extended lifetimes.
Active	1.0000	1.000 $\times$ Active	0	Baseline reference. All other bands are measured against this.
Hot	~ 985.1	$\sim 0.00102 \times$ Active	-3.0	Three orders of magnitude shorter. Compression kills lifetime fast.
Critical	$\sim 9.998 \times 10^7$	$\sim 1.0 \times 10^{-8} \times$ Active	-8.0	Eight orders of magnitude shorter. Critical-band structures are transient at every tier.

Section 8: Key Results and Patterns

Six findings emerge directly from the computations in Section 6 and 7. These are not interpretations added after the fact — they fall directly out of the formula structure.

Finding 1: Ghost and Active bands are nearly indistinguishable in lifetime at every tier.

The ratio Ghost/Active is approximately 1.0002 — a 0.02% difference. At every scale from the Planck floor to the observable universe, a near-void structure survives almost exactly as long as an equilibrium structure at the same D_r offset. This is because $D_r = 0.001$ gives $|D_r - 1| = 0.999$, which is nearly the same as the Active band's $|D_r - 1| = 1.000$, and the Ghost band's low beta (0.25) further suppresses any difference. The physical meaning: underdense voids are not fragile. They are slow, stable, long-lived structures — they simply disperse passively rather than collapse. The universe's vast empty spaces are as structurally durable as its densest equilibrium structures.

Finding 2: The Hot band cuts lifetime by three orders of magnitude at every tier.

Hot/Active approximately equals 0.00102 — a factor of 985 reduction in lifetime, independent of tier. A structure pushed from Active to Hot band loses 99.9% of its lifetime. This is the regime of stellar compression events, high-energy particle interactions, and geological fault-zone dynamics. Three orders of magnitude is a severe but not catastrophic penalty — structures in the Hot band are short-lived but not instantaneous, which is why neutron stars, compressed stellar cores, and dense particle jets can be observed at all. They have enough time to radiate, evolve, and reorganize before their frequency lock dissolves.

Finding 3: The Critical band cuts lifetime by eight orders of magnitude at every tier.

Critical/Active approximately equals 10^{-8} — a factor of 100 million reduction. Critical-band structures are transient on every scale — from T2 (10^{-51} s) to T41 (22 years). This is the pre-collapse zone without exception. Structures here are either on their way to exploding back to lower bands or collapsing toward UBL = 1. There is no stable Critical-band structure anywhere in the tier hierarchy. The Critical band is not a habitat — it is a transit corridor to collapse or explosion.

Finding 4: Lifetime scales with tier prime via $p^{1.618}$.

Moving from T2 ($p = 2$) to T41 ($p = 41$) adds 62 orders of magnitude to the baseline lifetime. The golden-ratio exponent amplifies the prime number into a large multiplier: $p = 2$ gives a multiplier of 3.07 while $p = 41$ gives a multiplier of 232. This is the substrate's built-in architecture: higher tiers are intrinsically more stable because they represent larger, more inertia-rich frequency locks. A galactic-scale structure has a vastly longer intrinsic coherence time than a particle-scale structure — not because it is qualitatively different in kind, but because the same substrate mechanism produces a $p^{1.618}$ scaling across the tier hierarchy.

Finding 5: $D_r = 1.0$ is the immortality threshold.

At perfect equilibrium, $|D_r - 1| = 0$ and the formula gives L approaching infinity. No physical structure perfectly achieves $D_r = 1.0$, but the formula's prediction is clear: the closer a structure sits to equilibrium, the longer it survives. Stable long-lived structures — protons (T5), main-sequence stars (T29), stable galaxies (T37) — are observed to sit extremely close to their tier's equilibrium density over long timescales. This is not coincidental in UST — it is the selection effect of the lifetime formula. Structures that form near $D_r = 1.0$ survive long enough to be observed. Structures that form far from $D_r = 1.0$ do not.

Finding 6: $UBL = 1$ is the collapse-bounce bridge.

The Critical-band lifetime at T41 is 22 years. If the observable universe ever entered a true Critical-band state — D_r exceeding 10,000 at the T41 scale — the entire observable cosmic structure would collapse within a human lifetime, and that collapse would be the bounce. The system would cross $UBL = 1$ and the substrate would reset toward UBL approaching 0, initiating the next expansion cycle. The brevity of that 22-year window is precisely the mechanism of the bounce: the collapse is too fast for any sub-structure at any tier below T41 to survive coherently. No frequency lock persists. No information is carried across. The slate is clean.

Section 9: Formula Limitations and Open Questions

9.1 Single-Tier Assignment

The formula assumes a structure occupies a single prime tier. Real structures span multiple tiers simultaneously. A human body is T17 cells inside a T19 organism living on a T23 planet orbiting a T29 star inside a T37 galaxy in a T41 universe. Each of those tier-level structures has its own D_r and its own lifetime formula result. A full UST description of a real object would sum contributions across all occupied tiers, with coupling terms between tiers. This paper isolates single-tier behavior to establish the baseline formulas. Cross-tier coupling is a separate problem that requires modeling how density perturbations at one tier propagate upward or downward into adjacent tier frequency locks. That work is not attempted here.

9.2 The $D_r = 1.0$ Singularity

The lifetime formula diverges at $D_r = 1.0$ — it predicts infinite lifetime at perfect equilibrium. No real structure achieves exact equilibrium. In practice, UST proposes that structures maintain a fluctuating D_r with a small epsilon offset from 1.0, giving finite but extremely long lifetimes. A proton, for example, might sit at $D_r = 1.0$ plus or minus epsilon where epsilon is on the order of 10^{-30} or smaller, producing a finite but astronomically long lifetime consistent with experimental lower bounds on proton stability. Quantifying epsilon for different structure types at each tier — deriving the minimum achievable D_r fluctuation amplitude from substrate dynamics — is a central open problem in UST. Until epsilon is derived, the formula gives order-of-magnitude results in the vicinity of $D_r = 1.0$ but not exact long-lifetime predictions for the most stable structures.

9.3 The Golden-Ratio Alpha Exponent

The tier-scaling exponent $\alpha = 1.618$ is geometrically motivated — it emerges from the self-similar recursive structure of the substrate's frequency hierarchy, where the ratio of consecutive resonance amplitudes converges to phi. This is the correct physical justification for the value, and it produces results consistent across all 13 tiers. However, it has not yet been derived from first principles starting from the substrate's fundamental field equations. A rigorous derivation would start from the substrate wave equation, show that the tier hierarchy emerges as the set of stable resonance modes, and demonstrate that the amplitude ratio between adjacent prime tiers converges to phi. That derivation is a future milestone for UST. Until it is done, $\alpha = 1.618$ is the geometrically-correct and empirically-consistent approximation, but it is not a formally proven result.

9.4 The Beta Exponent Values

The beta values by band — 0.25, 0.50, 1.00, 1.50, 2.00 — follow a perfectly regular arithmetic progression with a step of 0.25 per band. This simplicity may be real: it could reflect a genuine linear progression in the substrate's pressure-density response sensitivity across density regimes. Or it could be a placeholder approximation that captures the right order-of-magnitude behavior while hiding finer structure. The paper treats these as the best current approximation. More detailed substrate pressure-density modeling — particularly at the transitions between bands — might reveal that the progression is not exactly arithmetic or that the step size varies between the underdense and overdense sides of equilibrium. This is particularly relevant at the Ghost-to-Quiet and Hot-to-Critical transitions, where the physical regime shifts from passive dispersion to active compression behavior.

9.5 Cross-Band Coupling

Interactions between structures at different density bands are not modeled in this paper. A Hot-band structure embedded in a Ghost-band environment — for example, a stellar core at $D_r = 100$ inside a diffuse nebula at $D_r = 0.001$ — would require a coupling term in the lifetime formula that accounts for the density gradient at the interface. The current formula treats each structure as isolated in its own density band. In reality, density gradients between a structure and its environment drive interface dynamics — pressure waves, evaporation flows, condensation fronts — that modify the effective lifetime at the boundary. Cross-band interaction modeling, including the derivation of a coupling term for the lifetime formula, is reserved for future work.

Section 10: Conclusion

This paper derived lifetime formulas for all 13 prime tiers of UST using the core equation $L(p, D_r) = \tau_{\text{base}}(p) \times p^{1.618} \text{ divided by } |D_r - 1| \text{ raised to } \beta(\text{band})$. The five density bands — Ghost, Quiet, Active, Hot, and Critical — capture the full range of substrate stress conditions from near-void dispersion to pre-collapse compression. The computations are complete, explicit, and consistent across all tiers.

The results span 62 orders of magnitude in lifetime. At the bottom, T2 Critical-band events last 10^{-51} seconds — instantaneous micro-collapses at the Planck-floor substrate lattice. At the top, T41 Active-band structures survive approximately 2.2 trillion years — the coherence time of the observable universe as a frequency lock on the substrate. Between those endpoints, every known class of physical structure — particles, nuclei, atoms, molecules, cells, organisms, planets, stars, stellar clusters, galaxies, and the cosmic web — falls exactly where its physical behavior demands it should.

The most important physical result is this: $D_r = 1.0$ is the structural immortality threshold. The lifetime formula diverges to infinity at perfect equilibrium. Long-lived stable structures in nature — protons, main-sequence stars, stable galaxies — are observed to sit close to $D_r = 1.0$ at their respective tiers, and UST's lifetime formula gives the precise reason why: proximity to equilibrium is what keeps a frequency lock alive. Moving away from that balance — in either direction, through underdensity or overdensity — is what kills it.

The Critical band is the collapse gateway to $UBL = 1$. No Critical-band structure survives long enough to stabilize into a new equilibrium — it either explodes back to lower density or collapses through $UBL = 1$ and is absorbed back into the substrate. At T41, that collapse takes 22 years. That 22-year window is the bounce mechanism itself.

The formula is scale-universal. The same mathematical structure — a baseline timescale amplified by the golden-ratio power of the tier prime, divided by the density deviation raised to a band-specific sensitivity exponent — governs particle lifetimes at 10^{-26} seconds, stellar oscillation periods at 7.7 minutes, and cosmic web persistence at 2.2 trillion years. UST's core claim is that scale is not fundamental — it is an emergent property of which resonance level the substrate is expressing. The formula shows that this claim produces quantitatively consistent results across 62 orders of magnitude of time.

Three open questions remain: the derivation of cross-tier coupling terms, the first-principles derivation of $\alpha = 1.618$ from the substrate wave equation, and the quantification of the epsilon offset that replaces the $D_r = 1.0$ singularity with a finite maximum lifetime for perfectly-balanced structures. These are the next problems.

Appendix: Notation and Symbol Reference

Every symbol used in this paper is listed below with its full definition. Symbols are listed in order of first appearance.

Symbol	Name / Description	Units / Range	First Appears
UBL	Universal Balance Law ratio. The fraction of the substrate's total energy capacity that is currently locked into coherent structures.	Dimensionless, 0 to 1	Section 2.2
E_organized	Energy currently locked into coherent, structured frequency patterns — all matter and bound fields.	Joules (or equivalent)	Section 2.2
E_total	Maximum substrate energy capacity — the total energy if every substrate mode were excited to its resonance peak.	Joules (or equivalent)	Section 2.2
D_r	Density ratio. The ratio of local mass-energy density to the critical density at the structure's prime tier. The primary measure of substrate stress.	Dimensionless, 0 to infinity	Section 2.3
rho_local	Measured local mass-energy density at the structure's location.	kg/m ³ or equivalent	Section 2.3
rho_crit(p)	Critical density threshold for prime tier p. The equilibrium density at which a tier-p structure's self-gravity exactly balances outward substrate pressure.	kg/m ³ or equivalent	Section 2.3

Symbol	Name / Description	Units / Range	First Appears
p	Prime number of the tier. Labels the resonance level of the substrate at which the structure is organized. Takes values: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41.	Dimensionless integer (prime)	Section 2.4
T2, T3, ... T41	Tier designations for each of the 13 prime tiers. The number following T is the prime number p of that tier.	Label (dimensionless)	Section 2.4
L(p, D _r)	Predicted lifetime of a coherent structure at prime tier p and density ratio D _r . The decay time constant of the frequency lock under the given density conditions.	Seconds	Section 4.1
tau _{base} (p)	Baseline lifetime constant for prime tier p. Derived as the light-crossing time across the tier's characteristic length scale. The fundamental minimum response time of the structure to density stress.	Seconds	Section 4.1
alpha	Tier-scaling exponent. Set to 1.618 (the golden ratio, phi) in UST geometry. Reflects the self-similar recursive frequency architecture of the substrate's tier hierarchy.	Dimensionless, fixed at 1.618	Section 4.1
D _r - 1	Absolute deviation of the density ratio from equilibrium. The key stress measure in the lifetime formula. Zero only at D _r = 1.0 (perfect equilibrium), where L diverges to infinity.	Dimensionless, non-negative	Section 4.1
beta(band)	Density sensitivity exponent. Determines how steeply lifetime falls as D _r - 1 increases. Takes band-specific values: Ghost = 0.25, Quiet = 0.50, Active = 1.00, Hot = 1.50, Critical = 2.00.	Dimensionless	Section 4.1
phi	The golden ratio. phi = 1.618033... — the limit of the ratio of consecutive Fibonacci numbers. Used as alpha in the lifetime formula. Emerges from the recursive packing geometry of substrate frequency modes.	Dimensionless, irrational constant	Section 4.2

Symbol	Name / Description	Units / Range	First Appears
L_scale(p)	Characteristic length scale of prime tier p. The physical size of the structures that form at tier p. Used to derive tau_base(p) via tau_base = L_scale / c.	Meters	Section 4.4
c	Speed of light in vacuum. Used to convert characteristic length scales to timescales for tau_base calculation.	3.0×10^8 m/s	Section 4.4
epsilon	The minimum achievable D_r offset from perfect equilibrium in a real physical structure. Replaces the D_r = 1.0 singularity with a finite maximum lifetime. Not yet quantified for specific structure types — an open problem in UST.	Dimensionless, very small positive number	Section 9.2

This document presents original theoretical framework content developed under Unified Substrate Theory. No external references are cited; all formulas, tier assignments, and physical interpretations are original UST derivations.